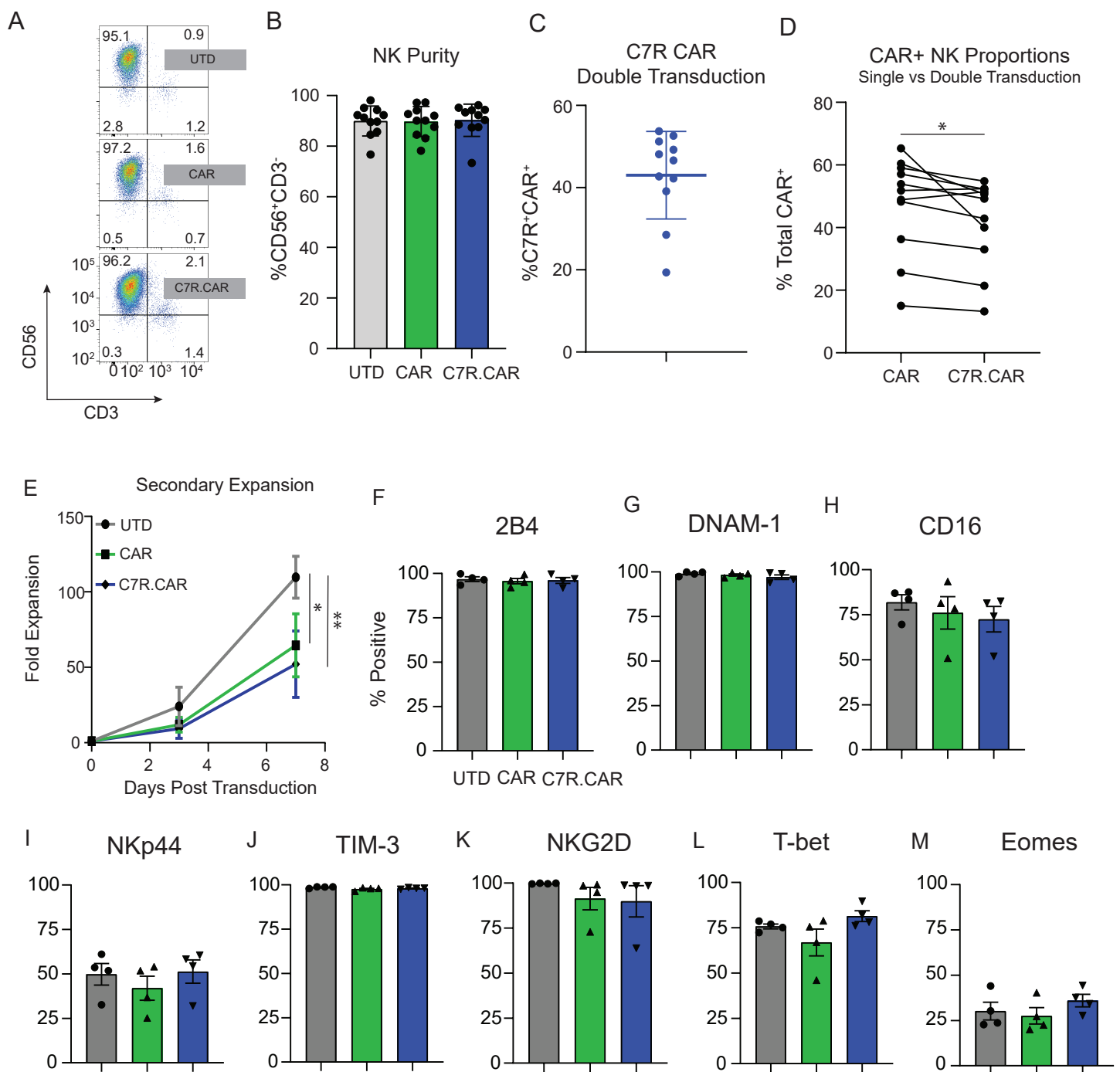
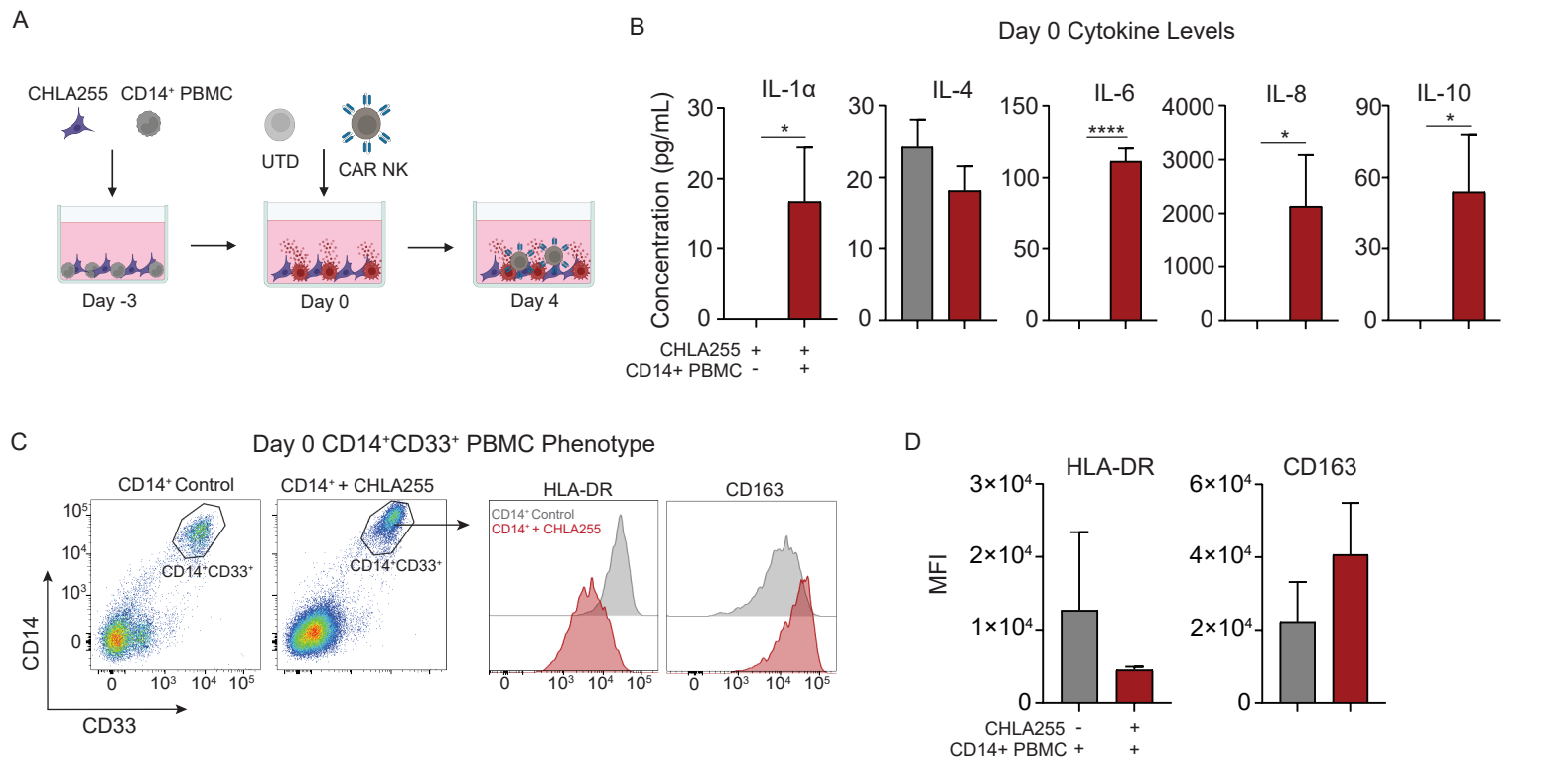
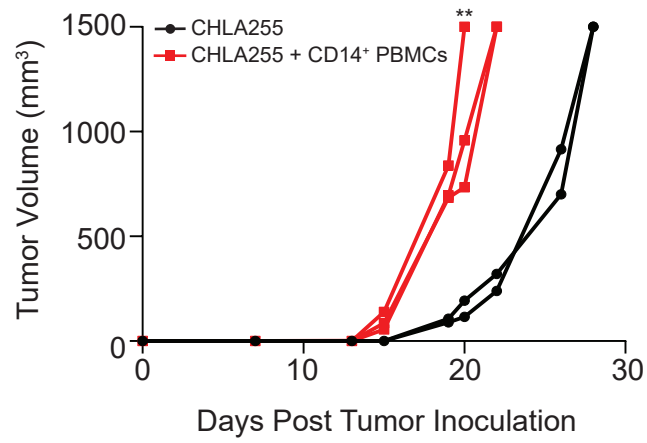




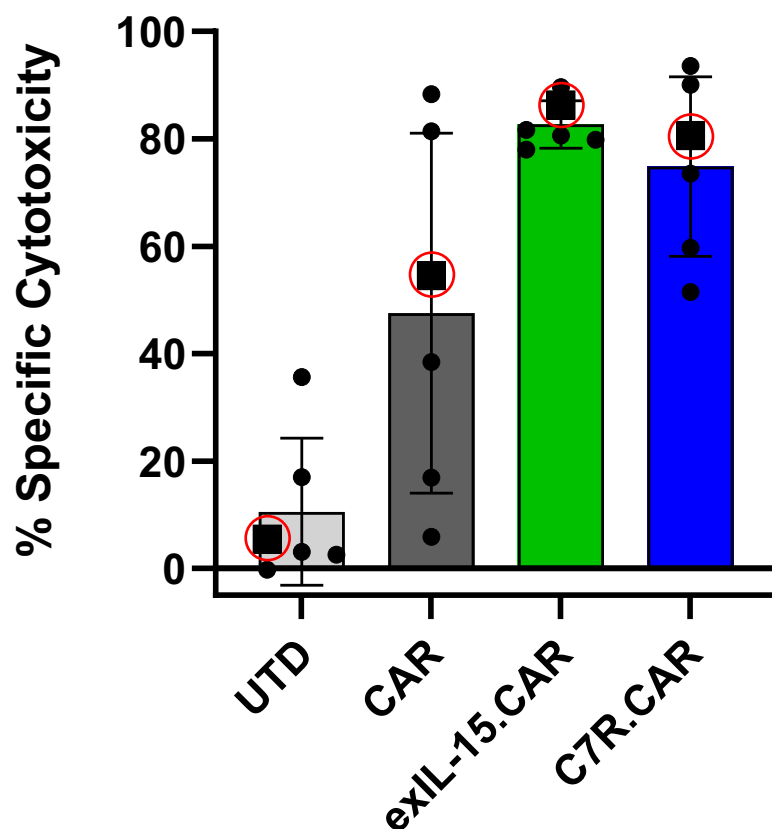
Supplementary Figure 1. C7R.CAR NK cells demonstrate no phenotypic differences and expand. (A) Representative flow plots of CD56⁺CD3⁻ purity after secondary expansion in indicated groups. **(B)** Pooled data from (A), n=11 independent donors. Data presented as mean ± stdev. **(C)** Double transduction rates in C7R.CAR NK cells, n=11 independent donors. Data presented as mean ± stdev. **(D)** Total CAR⁺ NK cells in single transduced CAR-NK cells or double transduced C7R.CAR NK cells. Data presented as paired individual points from independent donors. p < 0.05 (two-tailed paired t-test). **(E)** Cells were cultured at a 1:1 ratio with irradiated feeder cells and 10 ng/mL IL-15 and 10 I.U./mL IL-2 and cultured for up to 7 days in a G-rex. **(F-M)** CAR-NK phenotype in respective conditions after 4 days of secondary expansion. Data is gated on respective population and presented as mean ± stdev, n=4 independent donors.



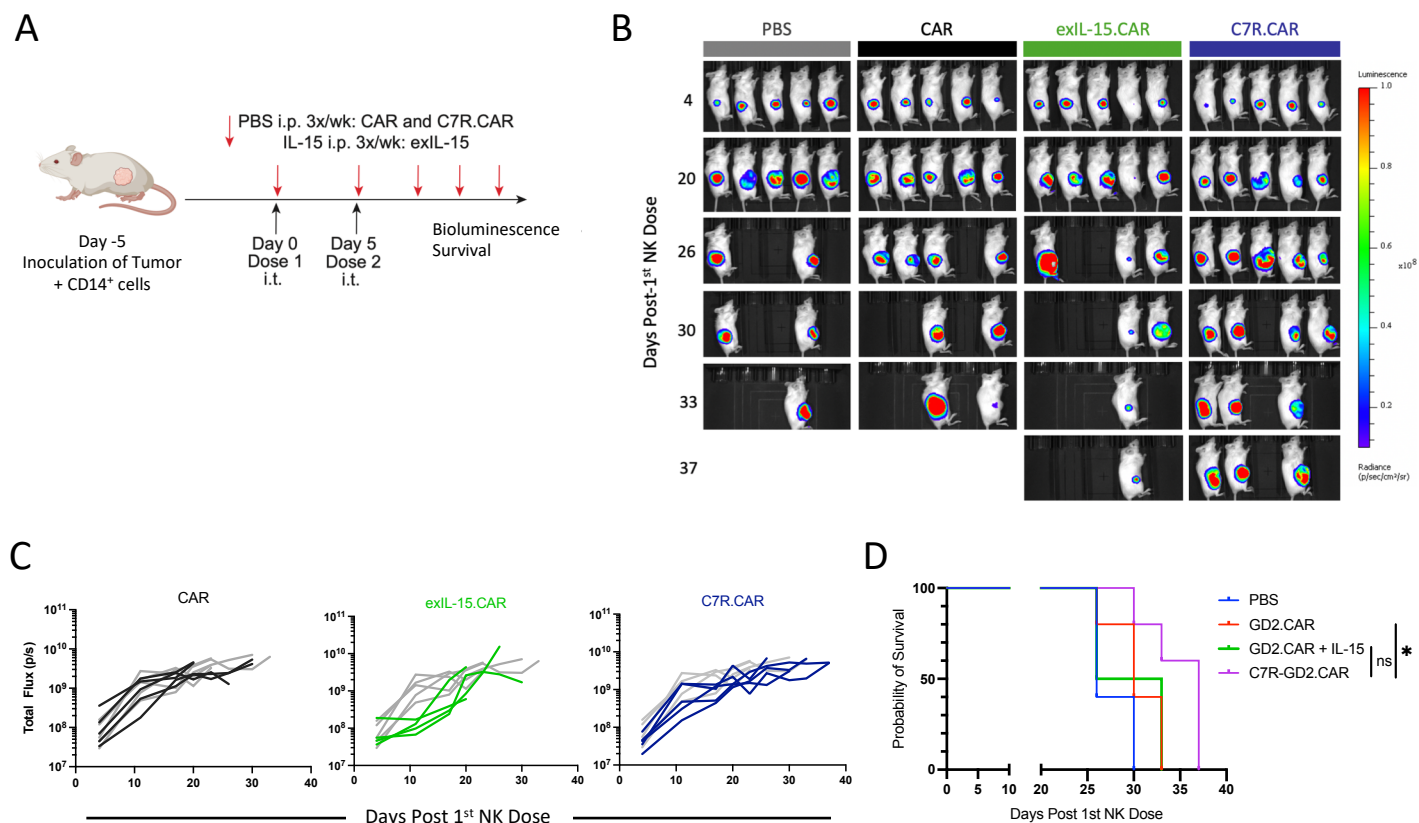
Supplementary Figure 2. *In vitro* TME mimic generates M2-like monocytes and inhibitory cytokines. (A) Schematic of TME generation. Briefly, 2.5×10^5 CHLA255 and 2.5×10^5 CD14⁺ PBMCs are co-cultured at a 1:1 ratio at day -3 prior to addition of 1×10^5 UTD or CAR NK cells on day 0. **(B)** Soluble levels of indicated cytokines at day 0 of tumor alone (gray) or tumor admixed with CD14⁺ PBMCs (red). Cytokines were measured via multiplex ELISA. **(C)** Gating strategy to determine CD14⁺CD33⁺ phenotype at day 0 of co-culture and representative histograms of HLA-DR and CD163 expression. **(D)** Quantitated HLA-DR and CD163 MFI values in CD14⁺ PBMCs cultured alone (gray) or cultured in the presence of CHLA255 (red). * $p < 0.05$, **** $p < 0.001$ (two-tailed t test, **B**). Graphs **B**, **D**, represent averages from different donors $\pm$ stdev ($n=3$, **B**, **D**)



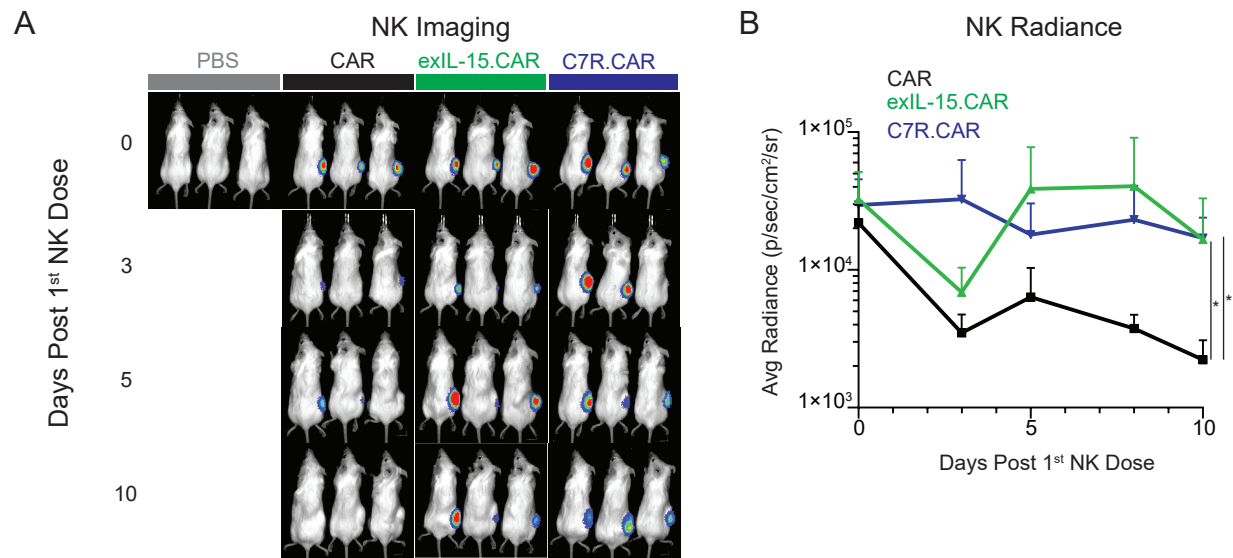
Supplementary Figure 3. CD14⁺ PBMCs accelerate tumor growth. (A) Tumor volume of indicated tumor xenograft condition. Two mice were used for CHLA255 and three mice were used for CHLA255 + CD14⁺ PBMCs. **p<0.01 (one-way ANOVA) at day 19 post tumor inoculation.



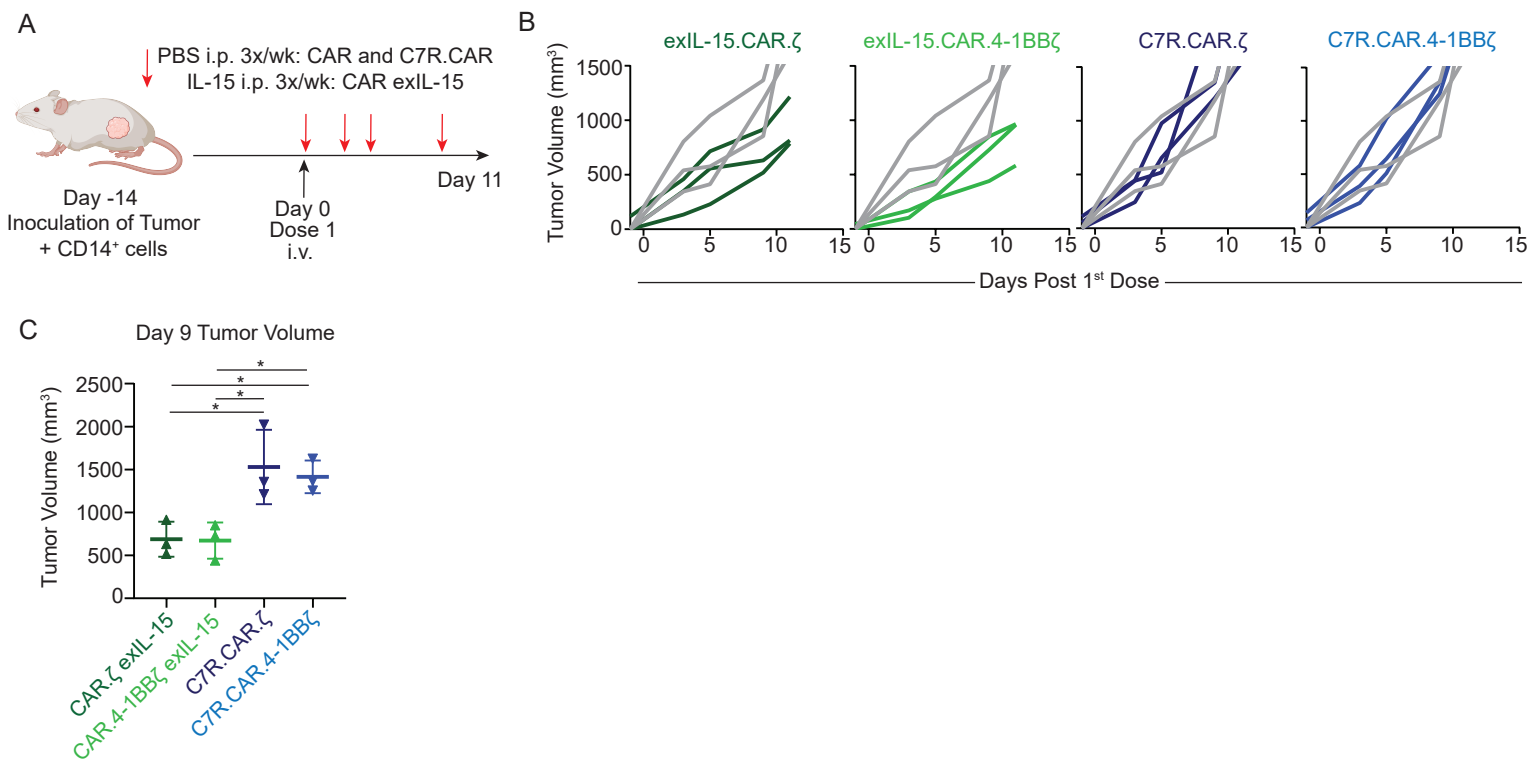
Supplementary Figure 4. Selection of *in vivo* CAR-NK cell donor. Modified version of Figure 2A, demonstrating cytotoxicity of individual donors. The squares from each condition represent the donor that was chosen for the *in vivo* assays. When compared to a group of independent donors used in these studies overall (n= 6), this donor (represented as the large black square highlighted by a red circle) exhibited low endogenous cytotoxicity and moderate cytotoxicity with CAR expression that was further enhanced with cytokine support. We used this criteria to select the *in vivo* donor for these studies because we wanted an NK donor in which most of the cytotoxicity was conferred by the CAR, not endogenous NK cells receptors such as KIR and NCR, thus truly assessing effects of cytokine modulation on CAR-specific activity in primary human NK cells.



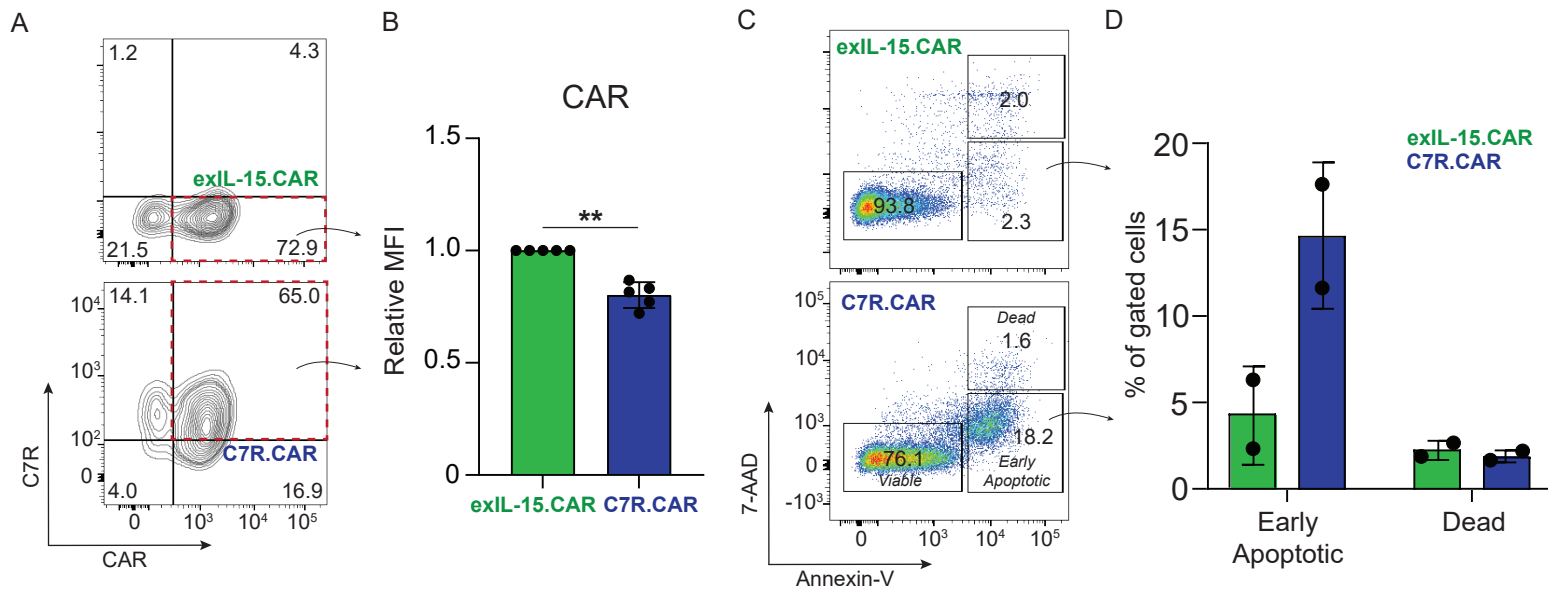
Supplementary Figure 5. C7R CAR NK given intratumorally do not control in vivo TME xenografts (A) *In vivo* schematic. Briefly, on day -5, 1×10^6 CHLA255 and 0.5×10^6 CD14⁺ PBMCs were implanted subcutaneously in the right flank of NSG mice. On day 0, 5×10^6 CAR NK cells were administered intratumorally (i.t.) followed by a second dose 5 days later. IL-15 (1 μ g/dose) was administered i.p. only in the exIL-15.CAR treated mice 3x/week. (B) Representative imaging of CHLA255 growth over time measured via bioluminescence. (C) Spider plots of individual mice quantified bioluminescent signal of CHLA255 over time from (B). (D) Survival analysis. $n=5$ mice per group; * $p = 0.04$ by Mantel-Cox log-rank test. Note: Given atypical starting tumor size of mouse #4 in exIL-15.CAR group, this mouse was removed from final statistical analysis.



Supplementary Figure 6. C7R Improves Survival in the TME. (A) Sequential bioluminescent imaging of NK cells. **(B)** Quantitated bioluminescent imaging of NK cells. Data is presented as the average±stdev between 3 mice from each group. $p^* < 0.05$ (one way ANOVA, **B**).



Supplementary Figure 7. Different signaling domains in CAR do not affect efficacy **(A)** In vivo schematic. Briefly, on day -14, CHLA255 and autologous CD14⁺ PBMCs are implanted subcutaneously in the right flank of NSG mice. On day 0, 5×10⁶ CAR NK cells were administered intravenously. IL-15 (1μg/dose) was administered i.p. only in the CAR exIL-15 treated mice 2-3x/week. **(B)** Spider plots of tumor volume in each indicated CAR NK condition overlayed with control PBS (gray) condition. Tumor volumes were determined by caliper measurement. **(C)** Respective tumor volumes on day 9 post NK administration. Data presented as average ± stdev, n=3 mice per group p* < 0.05 (one way ANOVA, **C**)



Supplementary Figure 8. C7R.CAR-NK cells demonstrate reduced surface CAR expression and higher proportions of early apoptotic cells. (A) Gating Strategy for quantifying relative MFI's of CAR and Annexin-V and 7-AAD staining. **(B)** Quantitated CAR MFI's. Data was normalized to exIL-15.CAR cell type. **(D)** Representative flow plot demonstrating Annexin-V and 7-AAD staining in gated cell populations indicated in A. **(E)** Quantitated data from D. Data presented as averages between independent donors $\pm$ stdev. (n=5, B, n=2, E). **p<0.01 (two-tailed paired t test, B).